ecbench round 3 run report: unrolled 32-bit inversion, walk states in the key ring

Run start: 2026-11-01T23:20Z (UTC). **Study:** `research/ecbench_walk_round3_20261102/`. **Class:** engineering; every child report identical apart from timing fields.

What ran

- **Binaries.** M3: `main` 8931f24b88 with this round's file reverted (sha256 4af2a84a89695625…); H: with it (4842ce99db727bd1…). Same tree otherwise.
- **Inputs.** The exported 32-bit child inputs (`data/input-32bit-*.json`, byte-identical under both binaries and round 2's) and the 20-bit `audit-ir` seq 0–2.
- **Instruments.** Callgrind (`--cache-sim=no --branch-sim=no`): marker split for the rho child, whole process for the IC child, `audit-ir` at 20 bits. Wall: the child's `solve_wall_ns`, interleaved M3 / H / M3, `taskset -c 3`, host load about 1.
- **Host.** Shared 4-vCPU cloud VM, Intel Xeon @ 2.10 GHz, no PMU; L0. valgrind 3.22.0, rustc 1.97.0.
- **Completed.** 2 binaries × (rho marker, rho whole, IC whole, 3 `audit-ir` children); 9 + 7 wall rounds. No failures. `verify --replay-all --exit-code` under H: 36/36 and 4/4 reproduced. `cargo test` (ic_boundary, workload, curve_id, ic_measurement): 77 passed, 0 failed; clippy clean apart from the pre-existing unknown-lint warning.

Measured values

Callgrind Ir	M3	H	change
32-bit `rho.negation` solve (34,156 GAE)	24,782,086	**22,368,502**	−9.7%
32-bit `rho.negation` whole child	27,311,486	**24,902,908**	−8.8%
32-bit `ic.pipeline` N=1024 whole child	776,961,624	**707,398,246**	−9.0%
20-bit `rho.negation` solve (1,706 GAE)	1,054,694	**993,690**	−5.8%
20-bit `ic.pipeline` N=32 solve (14,101 GAE)	12,399,287	**11,647,495**	−6.1%
20-bit `ic.pipeline` N=45 solve (10,632 GAE)	8,547,378	**8,026,386**	−6.1%

Wall (median, descriptive): 32-bit rho 6.04 ⟶ 5.98 ms (A/A 6.46 ms); 32-bit IC 287.6 ⟶ 284.0 ms (A/A 263.2 ms). No difference resolved: the A/A spread exceeds it.

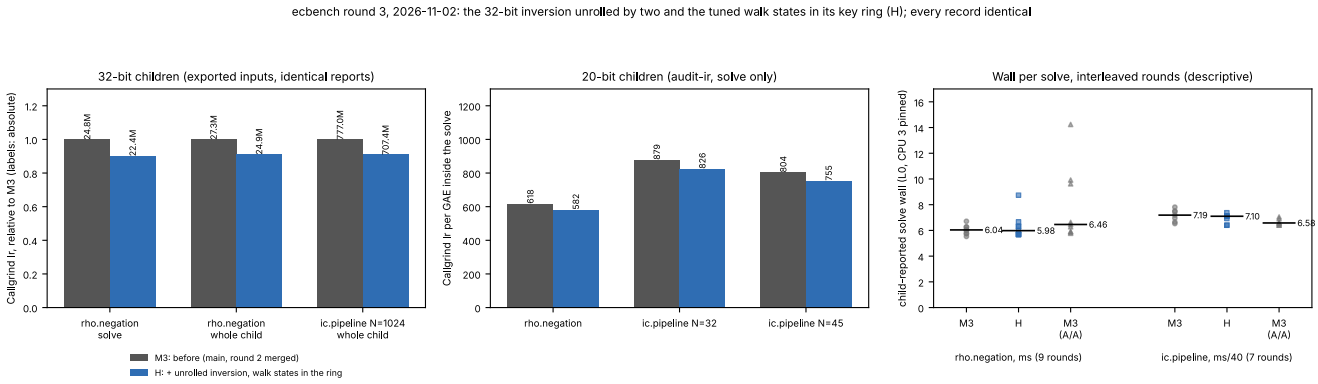


Figure 1: Recorded values only (`data/ir_32bit.tsv`, `data/ir_20bit.tsv`, `data/wall_rounds_32bit.tsv`).

Raw records

`data/ir_32bit.tsv`, `data/ir_20bit.tsv`, `data/wall_rounds_32bit.tsv`, `data/input-32bit-rho-negation.json`, `data/input-32bit-ic-pipeline.json`.